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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Machine Learning (course)

## Course outline

## About NPTEL ()

## How does an NPTEL online course work? ()

## Week 0 ()

## Week 1 ()

## Week 2 ()

☐ Linear Regression (unit? unit=32&lesson=33)

☐ Multivariate Regression (unit? unit=32&lesson=34)

☐ Subset Selection 1 (unit? unit=32&lesson=35)

## Week 2: Assignment 2 (Non Graded)

Assignment not submitted

**Note : This assignment is only for practice purpose and it will not be counted towards the Final score.**

1) Let  $A^{m \times n}$  be a matrix of real numbers. The matrix  $AA^T$  has an eigenvector  $x$  with eigenvalue  $b$ . Then the eigenvector  $y$  of  $A^T A$  which has eigenvalue  $b$  is equal to **1 point**

- ☐  $x^T A$   
☐  $x^T A$   
☐  $x$   
☐ Cannot be described in terms of  $x$

2) Let  $A^{n \times n}$  be a row stochastic matrix - in other words, all elements are non-negative and the sum of elements in every row is 1. Let  $b$  be an eigenvalue of  $A$ . Which of the following is true? **1 point**

- ☐  $|b| > 1$   
☐  $|b| \leq 1$   
☐  $|b| \geq 1$   
☐  $|b| < 1$

3) Let  $u$  be a  $n \times 1$  vector, such that  $u^T u = 1$ . Let  $I$  be the  $n \times n$  identity matrix. The  $n \times n$  matrix  $A$  is given by  $(I - kuu^T)$ , where  $k$  is a real constant.  $u$  itself is an eigenvector of  $A$ , with eigenvalue  $-1$ . What is the value of  $k$ ? **1 point**

- ☐ -2  
☐ -1  
☐ 2

- ☐ Subset Selection 2 (unit? unit=32&lesson=36)
- ☐ Shrinkage Methods (unit? unit=32&lesson=37)
- ☐ Principal Components Regression (unit? unit=32&lesson=38)
- ☐ Partial Least Squares (unit? unit=32&lesson=39)
- ☐ **Practice: Week 2: Assignment 2 (Non Graded) (assessment? name=265)**

☒ Quiz: Week 2: Assignment 2 (assessment? name=278)

☐ Week 2 Feedback Form : Introduction To Machine Learning (unit? unit=32&lesson=280)

☒ Week 2: Solution (unit? unit=32&lesson=209)

**Week 3 ()**

**Week 4 ()**

**Week 5 ()**

**Week 6 ()**

☐ 0

4) Which of the following are true for any  $m \times n$  matrix  $A$  of real numbers.

**1 point**

- ☐ The rowspace of  $A$  is the same as the column space of  $A^T$
- ☐ The rowspace of  $A$  is the same as the rowspace of  $A^T$
- ☐ The eigenvectors of  $AA^T$  are the same as the eigenvectors of  $A^T A$
- ☐ The **non-zero** eigenvalues of  $AA^T$  are the same as the **non-zero** eigenvalues of  $A^T A$

5) Consider the following 4 training examples

**1 point**

- $x = -1, y = 0.0319$
- $x = 0, y = 0.8692$
- $x = 1, y = 1.9566$
- $x = 2, y = 3.0343$

We want to learn a function  $f(x) = ax + b$  which is parametrized by  $(a, b)$ . Using squared error the loss function, which of the following parameters would you use to model this function.

- ☐ (1, 1)
- ☐ (1, 2)
- ☐ (2, 1)
- ☐ (2, 2)

**Check Answers and Submit**



[Week 7 \(\)](#)[Week 8 \(\)](#)[Week 9 \(\)](#)[Week 10 \(\)](#)[Week 11 \(\)](#)[Week 12 \(\)](#)[Text  
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